

LILLVIKEN-ROPARUDDEN V, Sweden

WMO: 021300

Lat: 66.4860N

Lon: 16.4173E

Elev: 481

StdP: 95.68

Time Zone: 1.00 (EUC)

Period: 01-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.3	(c) -29.5	(d) -35.0	(e) 0.1	(f) -32.3	(g) -32.1	(h) 0.2	(i) -29.5	(j) 15.9	(k) -0.2	(l) 14.0	(m) -2.1	(n) 1.3	(o) 320	(p) 0.810

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.3	21.5	12.5	19.6	11.7	17.6	11.2	14.8	17.6	13.8	16.4	12.9	15.5	3.7	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.9	10.5	15.1	12.8	9.8	14.2	11.8	9.1	13.7	43.1	17.6	40.2	16.7	37.7	15.6	16.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 14.3	(b) 12.1	(c) 10.4	(e) -35.8	(f) 24.4	(g) 2.5	(h) 2.3	(i) -37.6	(j) 26.1	(k) -39.0	(l) 27.4	(m) -40.4	(n) 28.7	(o) -42.2	(p) 30.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.2	-12.2	-11.0	-7.0	-1.4	4.3	9.3	12.8	11.6	7.3	1.9	-4.2	-9.1		
	DBStd	10.16	8.07	8.50	6.12	4.25	3.27	3.41	2.56	2.66	3.01	3.82	5.67	8.62		
	HDD10.0	3759	688	588	528	342	181	54	4	13	90	251	424	593		
	HDD18.3	6602	947	822	787	592	436	270	173	208	332	509	674	852		
	CDD10.0	198	0	0	0	0	3	34	91	63	8	0	0	0		
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0		
	CDH23.3	12	0	0	0	0	0	4	7	1	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	4.1	3.6	4.1	4.4	4.3	4.2	4.7	4.1	3.6	4.3	4.5	3.7	3.9		
(15) Precipitation	PrecAvg	669	59	45	42	31	39	63	91	77	65	55	49	55		
	PrecMax	878	92	101	72	58	65	105	151	162	136	90	95	112		
	PrecMin	484	19	11	14	11	11	12	27	30	19	26	16	22		
	PrecStd	104	23	24	17	13	17	30	37	39	30	17	23	22		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.7	4.5	5.8	9.6	20.4	23.3	24.2	23.1	16.7	10.3	7.0	5.3		
		MCWB	2.0	1.6	2.2	3.7	11.7	10.8	13.4	13.3	10.2	6.9	4.5	2.2		
	2%	DB	3.1	3.3	3.9	7.2	15.4	20.0	21.9	20.5	14.1	9.2	5.8	4.2		
		MCWB	0.6	0.9	1.4	3.1	8.1	10.3	13.0	13.0	10.2	6.2	3.3	1.5		
	5%	DB	1.3	2.1	2.7	5.7	12.4	17.7	20.1	18.2	12.7	8.2	4.1	2.8		
		MCWB	-0.5	-0.2	0.4	2.2	6.2	10.1	12.2	12.0	9.5	5.6	1.9	0.3		
	10%	DB	-1.4	0.4	1.5	4.4	9.9	15.8	18.0	16.3	11.6	7.3	2.4	1.2		
		MCWB	-2.7	-1.3	-0.6	1.4	5.0	9.8	11.8	11.7	8.8	5.0	0.7	-0.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.3	2.3	2.6	4.9	12.4	14.3	16.0	15.8	12.9	8.0	5.7	3.0		
		MCDB	4.5	3.8	4.4	8.1	20.7	16.6	19.2	18.6	14.2	9.6	6.3	4.8		
	2%	WB	0.9	1.1	1.5	3.7	9.1	12.9	15.1	14.6	11.2	6.8	3.8	1.6		
		MCDB	2.9	3.2	3.6	6.2	13.6	15.6	18.4	17.1	12.6	8.4	5.3	3.8		
	5%	WB	-0.6	-0.1	0.6	2.7	7.2	11.5	14.2	13.6	10.4	6.0	2.0	0.6		
		MCDB	1.1	1.7	2.6	5.1	10.8	15.3	16.7	16.3	11.9	7.9	3.9	2.5		
	10%	WB	-2.7	-1.3	-0.5	1.8	5.6	10.1	13.2	12.7	9.5	5.1	0.9	-0.5		
		MCDB	-1.4	0.4	1.2	4.2	8.9	14.8	15.8	15.0	11.2	7.0	2.0	1.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.1	9.0	9.1	7.9	7.1	7.2	7.3	6.7	5.3	4.0	4.7	7.1		
		MCDBR	5.9	8.2	7.8	8.5	13.0	11.6	11.7	10.6	7.2	4.4	5.1	6.3		
	5% WB	MCWBR	4.9	7.0	6.0	5.0	6.3	4.4	4.4	4.6	4.4	3.2	3.8	5.0		
		MCWBR	6.1	8.3	7.8	7.1	10.9	9.2	7.9	8.5	5.8	4.1	5.1	6.8		
(40) Clear-Sky Solar Irradiance	taub		0.175	0.242	0.280	0.306	0.320	0.323	0.333	0.319	0.291	0.262	0.180	0.178		
		taud	1.921	2.140	2.190	2.220	2.356	2.484	2.483	2.503	2.562	2.332	1.964	1.930		
	Ebn at Noon		242	620	779	852	881	889	865	843	794	601	221	22		
		Edh at Noon	15	51	82	104	102	92	90	79	59	42	13	3		
(44) All-Sky Solar Radiation	RadAvg		0.07	0.53	1.98	3.85	4.93	5.04	4.62	3.44	1.82	0.64	0.14	0.01		
	RadStd		0.00	0.09	0.15	0.23	0.41	0.36	0.54	0.35	0.20	0.10	0.01	0.00		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page